Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Hydrogen production from the water-gas shift reaction using Pt single atom catalysts**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), **Hydrogen (X)**, Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Chemical Technologies		**Energy & Resource Technologies**	
	Medium Category	Small Category	Medium Category	Small Category
	Petrochemicals & Catalysis	**Catalysis application technologies**	**Renewable Energy**	**Waste-to-energy tech**
Concept of Proposed Technology	Core Concept The water–gas shift (WGS) reaction ($CO + H_2O \rightarrow CO_2 + H_2$, $\Delta H° = -41.1$ kJ mol^{-1}) is a critical process for hydrogen production and CO removal in reforming-derived syngas streams. Hydrogen is a central energy carrier for decarbonizing chemical manufacturing, refining, and transportation, and its large-scale production relies heavily on reforming-derived synthesis gas streams that contain significant amounts of carbon monoxide (CO).[1,2] The WGS reaction is essential for maximizing hydrogen yield and lowering CO concentrations to levels compatible with downstream separation.[1] The novelty in the proposed technology is based on ligand-coordinated Pt single-atom catalysts (SACs) supported on shape-controlled anatase TiO_2 nanostructures for highly efficient hydrogen production via the WGS reaction. By loading isolated Pt atoms through tailored metal–ligand–support interactions, this strategy enhances precious metal utilization efficiency while enabling precise control			

	over the electronic structure and local coordination of the active sites. Preliminary studies have demonstrated the effectiveness of the catalysts and the promise that they have for improving hydrogen production. <u>Main Functions</u> The proposed catalyst system is designed to promote efficient low-temperature conversion of CO and H_2O to H_2 and CO_2, thereby enhancing hydrogen production while lowering CO levels. Isolated Pt atoms function as highly active sites for CO activation; the single-atom character ensures a high-level of reaction selectivity because of the uniformity of the active sites. The reducible TiO_2 support and coordinated ligands facilitate water activation and oxygen transfer via cooperative effects.[3] The TiO_2 support are synthesized nanocrystals with specific shapes and faceting, which maximizes the efficacy and durability of the catalyst. The ligand structure stabilizes Pt atoms against sintering and deactivation under steam-rich conditions. Together, these attributes result in high activity, excellent selectivity, and durable performance at low precious metal loadings.[3-5] <u>Potential Uses</u> The developed Pt-SAC technology has broad applicability in hydrogen production and purification platforms, particularly reforming-based processes utilizing fossil or renewable feedstocks.[1-5] Potential uses include integration into industrial hydrogen plants, pre-combustion carbon capture systems, and fuel processor units for fuel cell applications, where low-temperature CO cleanup is critical.[1-6] The high utilization efficiency of Pt and the enhanced catalyst stability provide a route to lowering catalyst costs and improving overall system durability. More broadly, the design strategies established here can be translated to other catalytic reactions involving CO and H_2O activation, supporting the development of advanced catalysts for sustainable energy and chemical production.[5-8]
Rationale for Joint R&D Project	This technology is at a stage where collaboration with an industrial R&D team will help advance the technology to large scale production of hydrogen. Our research group has developed expertise in this novel catalyst platform. We have demonstrated the function of the catalyst under reaction conditions and done extensive characterization of prototype catalysts before, during, and after reaction. At this stage, it is important to connect with an industrial partner to examine necessary refinements of the catalyst design and catalyst synthesis procedure that would make it more straightforward to scale-up the catalyst. Collaboration with a Korean company will provide essential sharing of science and technology. We anticipate that this cooperation could also facilitate the discovery of related catalyst systems that would further extend the benefits of the joint project.
Key Words	1. Single-atom heterogeneous catalyst 2. Metal-ligand coordination 3. Water-gas shift reaction 4. Hydrogen production 5. Industrial catalysis

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (2)	End of the Project	Level (6)
R&D Objectives	We aim to develop a single-atom heterogeneous catalyst consisting of Pt single-atoms, stabilized by organic ligands on reduced TiO_2 nanoparticle supports. These high surface area powders have exhibited remarkable reaction selectivity for hydrogen production.			
R&D Target Specifications	The target in this project is to produce large scale catalyst beds that maintain: (1) reaction selectivity above 90% (2) high levels of reaction activity (turn over frequency above 0.50 s^{-1}) and (3) performance for 3 weeks at operating temperature and pressure.			
R&D Plan	Building on our previous demonstration that ligand-stabilized Pt single-atom catalysts supported on high–surface-area oxides deliver high activity and selectivity in the WGS reaction, this work will systematically develop these catalysts toward hydrogen production rates relevant to industrial applications. The research plan comprises several key stages of development: ▪ Catalyst system development and scale-relevant synthesis The first phase will focus on develop a scalable catalyst platform featuring ligand-stabilized Pt single atoms dispersed on high surface area anatase TiO_2 supports. Shape-controlled TiO_2 nanostructures, including nanorods, nanosheets, nano-octahedra, and nanotubes, will be prepared using alkali-assisted hydrothermal and anion-mediated synthesis routes suitable for gram-scale production. Control over support morphology enables systematic tuning of exposed facets, defect concentrations, and metal–support interactions. Pt single atoms will be incorporated using a metal-ligand single-site approach that allows high Pt loadings while preserving atomic dispersion. Synthetic parameters, including precursor chemistry, ligand identity, and deposition conditions, will be optimized to ensure reproducibility and scalability. Single-atom catalyst sites have a high sensitivity to their local environment, much more so than metal nanoparticle catalysts. Therefore, the titania nanostructure variation will provide an opportunity to refine and improve the catalysts in activity, selectivity, and stability. Initial studies have shown differences in the catalyst activity based on the nanostructure shape and this can be leveraged for improvement of the catalyst. ▪ Ligand design and Metal–Ligand–Support Optimization A key aspect of the study involves the rational design of the ligand environment to stabilize isolated Pt atoms while maintaining access to catalytic sites. Ligand types and binding configurations will be systematically varied to tune Pt oxidation state, coordination geometry, and electronic structure. The strength of metal–ligand interactions will be optimized to provide resistance to sintering while ensuring effective CO adsorption and activation. The interplay between ligand coordination and TiO_2 surface properties will be investigated to reveal cooperative effects that enhance water activation and oxygen transfer, establishing clear structure–activity relationships for catalytic performance. We have identified several important factors in ligand design: strong electron-acceptor character to stabilize metal cation, planar geometry or other optimization for support interaction, and bidentate binding pocket for metal binding stability. Withing that space, there is an enormous field of ligands that can be explored. These can be used to tune the chemical state of the metal center. We have found that phenanthroline based ligands with ketone groups are very effective in stabilizing Pt SACs for WGS reaction. There are opportunities to explore in addition of other functional groups to			

further strengthen support interaction and also to consider the interplay of the ligand and support in defining the character of the metal, especially during steps of the reaction in which the support is an active redox partner.

- Advanced characterization and mechanistic studies

The structural and electronic features of the Pt-SACs will be examined using advanced techniques such as Transmission Electron Microscopy (TEM, to be conducted at RPI), aberration-corrected High-Angle Annular Dark-Field Scanning Transmission Electron Microscopy (HAADF-STEM, to be conducted at Brookhaven National Laboratory or other partner facilities), X-ray photoelectron spectroscopy (XPS, at RPI), and X-ray absorption spectroscopy (XAS, at Brookhaven or Argonne National Laboratory). In our lab, we have capability for diffuse reflectance infrared Fourier transform spectroscopy (DRIFTS), which can be conducted using CO titration before or after reaction to characterize metal centers. We can also run DRIFTS during the reaction to examine in situ catalyst behavior. These characterizations will confirm atomic dispersion, Pt coordination environments, and the stability of metal–ligand–support interactions under reaction conditions. By combining structural characterization with kinetic studies, the dominant WGS reaction pathways and rate-limiting steps, providing mechanistic insight into the cooperative function of isolated Pt atoms and TiO_2 supports.

- Catalytic Performance Evaluation and Stability Testing

Catalyst performance will be assessed under relevant WGS conditions, with emphasis on low-temperature activity, hydrogen yield, and CO conversion efficiency. Shape-controlled supports and ligand- modified Pt-SACs will be compared to conventional Pt nanoparticle catalysts to evaluate gains in intrinsic activity and Pt utilization. Long-term stability tests will be conducted under steam-rich conditions and high space velocities will determine resistance to sintering, ligand degradation, and support restructuring, while post-reaction characterization will be used to correlate performance decline to structural changes. Our reactor system is a flow reactor that allows for mixing input gases using mass flow controllers. This allows us to precisely define the reaction gas conditions. After flowing over the catalyst, the gas mixture is characterized by mass spectrometry or by a gas chromatograph. This allows for real time characterization of the gas and assessment of the catalyst performance.

- Validation under scale-relevant conditions and translation toward practical applications

The top-performing catalyst formulations will be tested under scale-relevant conditions to assess their suitability for industrial applications. Key performance indicators, including turnover frequency, hydrogen yield, and durability, will be benchmarked against current state-of-the-art WGS catalysts. This study will provide practical design principles for atom-efficient WGS catalysts and establish a scalable materials platform for hydrogen production, ultimately reducing noble metal usage, improving catalyst lifetime, and enabling more cost-effective and reliable hydrogen generation systems.

Research Approach

To advance single-atom catalysts (SACs) toward industrially relevant hydrogen production, this project will systematically address catalyst scalability, activity optimization, and long-term stability under water–gas shift (WGS) reaction conditions.

Building on our prior demonstration that ligand-stabilized Pt-SACs, the proposed research integrates support morphology control, metal–ligand coordination tuning, and mechanistic evaluation to establish robust structure performance relationships.[3,4] The approach integrates rational catalyst synthesis, advanced characterization, and performance evaluation to optimize activity, stability, and scalability. It includes:

- Morphology-controlled TiO_2 supports for single-atom stabilization.
- Deposition and stabilization of single-atom catalysts enabling measurable catalytic activity

under scale up relevant space velocities.

- Role of ligand coordination and metal–ligand optimization.
- Catalytic performance evaluation on activity, stability, and scalability.
- Structure–activity–stability correlation: By integrating morphology-controlled supports, ligand coordination tuning, and rigorous catalytic evaluation.
- Establish clear correlations between: TiO_2 surface structure and Pt coordination environment, metal–ligand interactions and intrinsic catalytic activity and catalyst architecture and long-term operational stability

The most promising formulations will be benchmarked against conventional Pt nanoparticles and tested under scale-relevant conditions to establish design principles for atom-efficient, stable, and industrially viable WGS catalysts.

Key Tasks

Key research tasks include:

- Synthesize and stabilize Pt single-atom catalysts on selected shape-controlled anatase TiO_2 supports
- Advanced structural and electronic characterization, demonstration of ligand-enabled control over single-atom reactivity.
- Enhance low-temperature WGS performance with improved durability.
- Investigate structure–activity relationships for WGS performance
- Optimize catalytic activity and durability under realistic WGS conditions
- Reaction kinetics and mechanistic studies
- Assess scalability and productivity at elevated space velocities
- Stability, durability, and scale-relevant validation

These studies will elucidate catalyst structural evolution under WGS conditions and establish design guidelines for atom-efficient, durable WGS catalysts for hydrogen production.

Methodologies

The proposed research will employ an integrated experimental methodology combining rational catalyst synthesis, advanced materials characterization, and catalytic performance evaluation to develop scalable and robust Pt-SACs for the WGS reaction. The first phase will focus on the synthesis of shape-controlled anatase TiO_2 supports, including nanorods, nanosheets, nano-octahedra, and nanotubes, using alkali-assisted hydrothermal and anion-directed methods. These nanostructures provide controlled exposure of crystallographic facets and defect environments, enabling precise tuning of metal–support interactions and stabilization of isolated Pt sites. Pt single atoms will be deposited using ligand-assisted strategies at controlled loadings to maintain atomic dispersion while increasing active-site density.

Advanced characterization techniques will be used to probe catalyst structure and stability, including aberration-corrected HAADF-STEM for direct visualization of isolated Pt atoms, X-ray photoelectron spectroscopy (XPS) and X-ray absorption spectroscopy (XAS) to determine Pt oxidation state and coordination environment with support morphology and defect chemistry.

Catalytic performance will be assessed using fixed-bed flow reactors under representative WGS conditions, focusing on low-temperature activity, hydrogen productivity, CO conversion, and selectivity. Kinetic studies will be performed to elucidate dominant reaction pathways and identify rate-determining steps. Extended stability tests under steam-rich and high space-velocity conditions will evaluate resistance to sintering and ligand degradation. Collectively, these approaches will establish structure–activity–stability relationships and inform the development of atom-efficient,

scalable, and industrially viable WGS catalysts.

The reproducibility and scale-up potential of the synthesis methods will be developed and validated at increased batch sizes. The resulting catalyst systems will be evaluated under scale-relevant WGS operating conditions to assess their industrial viability.

Expected Milestones

The proposed research is expected to deliver fundamental insights and practical advances that bridge the gap between laboratory-scale SACs and industrially relevant hydrogen production via the water–gas shift reaction. Therefore, expected outcomes will be:

- Reproducible synthesis protocols, atomically dispersed Pt-SACs on TiO_2 morphologies, baseline structural characterization.
- Identification of optimal TiO_2 support morphologies for SAC stabilization and activity.
- Optimized ligand–Pt–TiO_2 architectures, structure–activity maps, identification of high-performance catalyst formulations.
- Investigate structure–activity–stability relationships for Pt-SACs supported TiO_2
- Validation of SAC productivity under industrially relevant conditions.
- Design principles for cost-effective and reliable hydrogen catalysts.
- Develop scale-relevant performance validation, catalyst design principles, technology translation framework.
- Provide stability benchmarks, durability data, scalable synthesis protocols.

Timeline Required to Achieve the Stated R&D Objectives

Phase I (Months 1–6): Platform Development, Materials Synthesis and characterization

This phase will focus on establishing a reproducible and scalable catalyst synthesis platform. Shape-controlled anatase TiO_2 nanostructures (nanorods, nanosheets, nano-octahedra, and nanotubes) will be synthesized and optimization of synthesis parameters. Ligand-stabilized Pt SACs will be prepared on these supports using a metal–ligand single-site strategy. Characterization will be performed, determine Pt coordination environments and metal–support interactions. Initial catalytic testing of WGS will be conducted.

Phase II (Months 6–12): Catalyst Platform Development, Ligand Optimization and Baseline Performance

This phase will systematically optimize the ligand environment and metal–ligand–support interactions. the ligand engineering and structure–activity optimization and tune Pt electronic structure, coordination geometry to improve Pt stabilization, enhance CO and H_2O activation. Advanced characterization techniques will be employed to monitor catalyst structure under reaction conditions.

Phase III (Months 12–18): Mechanistic Studies and Performance Evaluation

Kinetic analysis and studies reaction mechanisms, dominant pathways, and rate-limiting steps will identify dominant WGS pathways and rate-limiting steps, enabling mechanistic differentiation between Pt-SACs and Pt nanoparticles. Concurrently, durability tests under steam-rich and elevated space-velocity conditions will guide improvements in catalyst lifetime and resistance to deactivation. Develop mechanistic models for WGS on Pt-SACs, kinetic parameters, validated structure–function relationships.

Phase IV (Months 18–24): Stability, Durability, and Scalability Assessment

This phase will focus on long-term durability tests and performance under scale-relevant

	operating steam-rich conditions and elevated space velocities to evaluate resistance to sintering, ligand degradation. Reproducibility and scale-up potential of synthesis methods will be validated at increased batch sizes. Performance metrics such as turnover frequency, hydrogen yield, CO conversion, and long-term stability will be compared against state-of-the-art WGS catalysts. Phase V (Months 24–36): Scale-Relevant Validation and Technology Translation In this phase, the most promising catalyst formulations will be evaluated at higher throughput and extended time-on-stream to assess industrial feasibility. Stability benchmarks, durability metrics, and scalable synthesis protocols will be established. The project will culminate in the delivery of a scalable Pt-SAC materials platform, validated structure–performance relationships, and practical design guidelines to support atom-efficient, industrially viable WGS catalysts for future hydrogen production technologies.

4. Role & Responsibility (R&R)

Expected R&R of Research Institute	Prof. Tait and Dr. Wanna have research laboratory space at RPI that will be used for experimental measurements of the catalysts. This includes facilities for catalyst synthesis and characterization, as well as flow reactor systems that are equipped with either mass spectrometry or gas chromatograph analyzers. Diffuse reflectance infrared Fourier transform spectroscopy (DRIFTS) measurements will also be performed in the lab in a reaction cell that allows analysis under reaction conditions. We also have access to shared characterization facilities at RPI, including X-ray Photoelectron Spectroscopy (XPS) and Transmission Electron Microscopy (TEM). EXAFS measurements will be conducted at Brookhaven National Laboratory, which is also located in New York State.
Expected R&R of Korean SME Partner	We propose that the Korean SME partner will provide consultation and advice on preparing the catalyst systems for real reactor conditions and for the scale-up of catalyst production. In the later stages of the project, it will be helpful to collaborate on running water-gas shift in a test reactor.

5. Expected Outcome/Impact

Expected Outcomes and Impact	Key outcomes of this project include: • Advancing understanding of single-atom catalysis • Development of highly selective water-gas shift catalyst with long-term stability • Creating new catalysts to advance hydrogen production industry • Develop new industrial catalysts that can scale up to large-scale production More broadly, this work will benefit the research community and broader society by: • Training new scientists in advanced catalysis research • Demonstrating connection between catalyst characterization and reaction efficacy • Creating new production methods for affordable hydrogen resources • Addressing global energy needs

6. References

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☐ Terms:

1) Industrial Technology Classification

Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization